BEAM DESIGN SUMMARY

Project Name	:	G+2 RB BEAM DESIGN @FLOOR 1
Client Name	:	Unassigned
Engineer Name	:	Unassigned
Design File	:	E:\MY PROJECT 4-1\STAAD\RCDC FILES\G+2 Project Plan-Beam-1-5.029 m.R1.rcdx
Analysis File	:	E:\MY PROJECT 4-1\STAAD\G+2 Project Plan.std
Analysis Last Modified	:	06-10-2026 00:54:36
Level Designed	:	5.029 m

Beam No		:	**B1**	
Group No		:	**G1**	
Analysis Reference(Member)	5.029 m	:	3	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	76.03	49.71	41.1	85.93	36.9	92.6
PtClc (%)	0.579	0.358	0.292	0.669	0.26	0.734
AstCalc (sqmm)	492.42	304.93	248.12	569.69	221.14	624.25
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	64.41	58.18	81.73
Asv Torsion (sqmm/m)	-	204.61	279.17
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	718.14
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 140

SFR : -

Beam No		:	**B2**

Group No		:	**G1**	
Analysis Reference(Member)	5.029 m	:	4	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	47.92	43.67	60.95	80.58	47.88	111.84
PtClc (%)	0.344	0.311	0.449	0.62	0.344	0.908
AstCalc (sqmm)	292.93	264.9	382.23	527.39	292.69	772.83
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	72.58	50.5	74.05
Asv Torsion (sqmm/m)	251.95	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B3**	
Group No		:	**G1**	
Analysis Reference(Member)	5.029 m	:	140	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	26.37	26.37	26.37	62.49	52.57	35.37
PtClc (%)	0.2	0.2	0.2	0.462	0.381	0.248
AstCalc (sqmm)	170.2	170.2	170.2	393.1	324.21	211.38
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	138.9	135.47	132.04
Asv Torsion (sqmm/m)	919.84	908.98	898.12
Asv Reqd (sqmm/m)	615.43	589.75	564.06
Asv Prv (sqmm/m)	957.52	914	914
Reinforcement	2L-T8 @ 105	2L-T8 @ 110	2L-T8 @ 110

SFR : -

Beam No		**:**	**B4**	
Group No		**:**	**G2**	
Analysis Reference(Member)	5.029 m	:	23	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	88.84	46.31	53.94	85.57	43.11	85.95
PtClc (%)	0.697	0.332	0.392	0.666	0.307	0.67
AstCalc (sqmm)	593.19	282.28	333.61	566.83	261.25	569.83
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	52.47	52.25	64.03
Asv Torsion (sqmm/m)	173.62	174.54	211.82
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B5**	
Group No		:	**G2**	
Analysis Reference(Member)	5.029 m	:	24	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	61.93	41.3	67.5	71.77	60.98	118.19
PtClc (%)	0.457	0.293	0.504	0.541	0.449	0.958
AstCalc (sqmm)	389.13	249.46	429.06	460.46	382.4	815.6
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	51.48	52.6	64.38
Asv Torsion (sqmm/m)	170.76	173.7	210.98
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No : **B6**

Group No		:	**G2**	
Analysis Reference(Member)	5.029 m	:	145	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	23.09	23.09	26.32	80.65	64.02	34.28
PtClc (%)	0.2	0.2	0.2	0.62	0.475	0.24
AstCalc (sqmm)	170.2	170.2	170.2	527.96	404.01	204.52
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	74.57	73.74	71.16
Asv Torsion (sqmm/m)	304.21	301.59	293.42
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B7**	
Group No		:	**G3**	
Analysis Reference(Member)	5.029 m	:	43	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	93.54	46.1	62.29	90.19	44.59	94.99
PtClc (%)	0.743	0.33	0.46	0.71	0.318	0.757
AstCalc (sqmm)	632.18	280.88	391.66	604.28	270.94	644.43
Ast Prv (sqmm)	942.48	942.48	942.48	942.48	942.48	942.48
Reinforcement	3-T20	3-T20	3-T20	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	56.97	53.94	66.1
Asv Torsion (sqmm/m)	194.19	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B8**	
Group No		:	**G3**	
Analysis Reference(Member)	5.029 m	:	44	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	71.77	43.4	73.09	79.07	61.52	122.13
PtClc (%)	0.541	0.309	0.553	0.606	0.454	0.99
AstCalc (sqmm)	460.46	263.13	470.27	515.69	386.25	842.19
Ast Prv (sqmm)	942.48	942.48	942.48	942.48	942.48	942.48
Reinforcement	3-T20	3-T20	3-T20	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	53.3	57.71	69.49
Asv Torsion (sqmm/m)	-	197.17	234.45
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	718.14
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 140

SFR : -

Beam No		:	**B9**	
Group No		:	**G3**	
Analysis Reference(Member)	5.029 m	:	150	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	25.73	25.73	25.73	70.56	57.53	36.85
PtClc (%)	0.2	0.2	0.2	0.531	0.421	0.259
AstCalc (sqmm)	170.2	170.2	170.2	451.53	358.3	220.81
Ast Prv (sqmm)	942.48	942.48	942.48	942.48	942.48	942.48
Reinforcement	3-T20	3-T20	3-T20	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	76.59	75.76	73.18
Asv Torsion (sqmm/m)	365.32	362.7	354.52
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No : **B10**

Group No		:	**G4**	
Analysis Reference(Member)	5.029 m	:	63	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	88.46	51.91	57.68	98.81	42.53	108.9
PtClc (%)	0.694	0.376	0.422	0.796	0.303	0.885
AstCalc (sqmm)	590.17	319.76	359.35	677.26	257.46	752.97
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	75.2	61.48	85.04
Asv Torsion (sqmm/m)	254.39	-	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B11**	
Group No		:	**G4**	
Analysis Reference(Member)	5.029 m	:	64	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	64.9	43.95	66.24	92.4	54.33	126.22
PtClc (%)	0.482	0.313	0.494	0.732	0.395	1.022
AstCalc (sqmm)	410.33	266.73	419.98	622.65	336.25	869.76
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	72.82	60.13	83.69
Asv Torsion (sqmm/m)	234.46	197.21	271.77
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 140

SFR : -

Beam No		**:**	**B12**	
Group No		**:**	**G4**	
Analysis Reference(Member)	5.029 m	:	155	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	13.67	13.67	21.36	72.79	54.74	21.42
PtClc (%)	0.2	0.2	0.2	0.55	0.398	0.2
AstCalc (sqmm)	170.2	170.2	170.2	468.09	339.07	170.2
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	96.46	95.69	102.4
Asv Torsion (sqmm/m)	493.58	491.13	561.91
Asv Reqd (sqmm/m)	297.75	291.97	408.38
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B13**	
Group No		:	**G5**	
Analysis Reference(Member)	5.029 m	:	136	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	19.5	21.83	13.91	12.53	4.64	14.44
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Shear Design

	Left	Mid	Right
Vut (kN)	38.68	23.71	42.74
Asv Torsion (sqmm/m)	195.11	137.35	207.68
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 140

SFR : -

Beam No		:	**B14**

Group No		:	**G5**	
Analysis Reference(Member)	5.029 m	:	137	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	9.53	15.96	13.9	13.56	13.37	30.92
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.216
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	183.4
Ast Prv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	34.33	16.42	36.03
Asv Torsion (sqmm/m)	165.79	66.08	158.25
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B15**	
Group No		:	**G5**	
Analysis Reference(Member)	5.029 m	:	158	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	18.16	18.16	18.16	39.73	35.05	28.75
PtClc (%)	0.2	0.2	0.2	0.281	0.246	0.2
AstCalc (sqmm)	170.2	170.2	170.2	239.26	209.38	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	87.61	84.18	80.75
Asv Torsion (sqmm/m)	557.84	547.22	536.6
Asv Reqd (sqmm/m)	372.24	346.55	320.86
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B16**	
Group No		:	**G6**	
Analysis Reference(Member)	5.029 m	:	123	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	17.13	17.13	17.13	28.36	56.22	74.06
PtClc (%)	0.2	0.2	0.2	0.2	0.41	0.561
AstCalc (sqmm)	170.2	170.2	170.2	170.2	349.26	477.53
Ast Prv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	84.86	91.63	96.7
Asv Torsion (sqmm/m)	540.3	561.72	577.77
Asv Reqd (sqmm/m)	254.81	261.55	299.51
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B17**	
Group No		:	**G6**	
Analysis Reference(Member)	5.029 m	:	108	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	39.66	34.05	43.42	104.56	35.89	81.87
PtClc (%)	0.281	0.239	0.309	0.85	0.252	0.632
AstCalc (sqmm)	238.82	203.03	263.24	723.7	214.69	537.46
Ast Prv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	72.95	45.96	67.88
Asv Torsion (sqmm/m)	238.5	153.08	234.28
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No : **B18**

Group No		:	**G6**	
Analysis Reference(Member)	5.029 m	:	96	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	26.43	33.27	32.02	93.41	23.31	80.94
PtClc (%)	0.2	0.233	0.224	0.742	0.2	0.623
AstCalc (sqmm)	170.2	198.12	190.24	631.08	170.2	530.17
Ast Prv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	68.75	38.33	63.65
Asv Torsion (sqmm/m)	223.63	127.35	207.32
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B19**	
Group No		:	**G6**	
Analysis Reference(Member)	5.029 m	:	84	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	27.78	43.5	44.46	100.92	25.64	83.5
PtClc (%)	0.2	0.31	0.317	0.818	0.2	0.647
AstCalc (sqmm)	170.2	263.75	270.04	695.78	170.2	550.29
Ast Prv (sqmm)	339.3	339.3	339.3	942.48	942.48	942.48
Reinforcement	3-T12	3-T12	3-T12	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	76.27	46.32	64.19
Asv Torsion (sqmm/m)	261.65	166.82	208.32
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		**:**	**B20**	
Group No		**:**	**G7**	
Analysis Reference(Member)	5.029 m	:	124	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	13.9	11.66	11.66	24.54	69.74	97.7
PtClc (%)	0.2	0.2	0.2	0.2	0.523	0.785
AstCalc (sqmm)	170.2	170.2	170.2	170.2	445.44	667.66
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	45.91	52.16	54.16
Asv Torsion (sqmm/m)	151.08	170.86	177.19
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B21**	
Group No		:	**G7**	
Analysis Reference(Member)	5.029 m	:	109	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	65.14	42.38	72.62	108.79	47.26	83.91
PtClc (%)	0.484	0.301	0.549	0.884	0.339	0.65
AstCalc (sqmm)	412.05	256.45	466.79	752.26	288.57	553.57
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	Mid	Right
Vut (kN)	66.14	48.85	54.46
Asv Torsion (sqmm/m)	214.14	159.4	177.56
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No : **B22**

Group No		:	**G7**	
Analysis Reference(Member)	5.029 m	:	97	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	46.77	34.83	52.18	90.81	33.81	71.18
PtClc (%)	0.335	0.244	0.378	0.716	0.237	0.536
AstCalc (sqmm)	285.33	207.98	321.62	609.39	201.53	456.11
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	56	36.76	48.29
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B23**	
Group No		:	**G7**	
Analysis Reference(Member)	5.029 m	:	85	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	43.94	44.37	63.47	90.11	31.37	73.78
PtClc (%)	0.313	0.317	0.47	0.709	0.219	0.559
AstCalc (sqmm)	266.67	269.47	400.12	603.66	186.18	475.45
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	59.24	40.18	47.6
Asv Torsion (sqmm/m)	192.96	132.63	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B24**	
Group No		:	**G8**	
Analysis Reference(Member)	5.029 m	:	125	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Cantilever Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	25.32	14.91	14.91	24	68.24	92.55
PtClc (%)	0.2	0.2	0.2	0.2	0.511	0.733
AstCalc (sqmm)	170.2	170.2	170.2	170.2	434.51	623.85
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	97.41	102.26	104.27

Asv Torsion (sqmm/m)	565.96	581.29	587.66
Asv Reqd (sqmm/m)	304.88	341.12	356.2
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B25**	
Group No		:	**G8**	
Analysis Reference(Member)	5.029 m	:	110	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	54.68	41.62	58.88	120.85	44.2	97.19
PtClc (%)	0.398	0.296	0.432	0.979	0.315	0.779
AstCalc (sqmm)	338.62	251.55	367.71	833.53	268.34	663.23
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	79.86	52.88	76.15
Asv Torsion (sqmm/m)	259.56	174.14	268.42
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 140

SFR : -

Beam No		:	**B26**
Group No		:	**G8**
Analysis Reference(Member)	5.029 m	:	98

Breadth	:	230	mm
Depth	:	430	mm
Concrete Grade	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover	:	30	mm
Side Clear Cover	:	30	mm
Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type	:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	36.74	42.05	46.2	110.12	31.58	93.81
PtClc (%)	0.259	0.299	0.331	0.895	0.22	0.746
AstCalc (sqmm)	220.09	254.32	281.57	761.22	187.5	634.44
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	78.21	47.79	67.54
Asv Torsion (sqmm/m)	267.53	171.26	-
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	365.6
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B27**	
Group No		:	**G8**	
Analysis Reference(Member)	5.029 m	:	86	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	41.9	52.99	61.28	116.45	33.11	101.11
PtClc (%)	0.298	0.384	0.452	0.945	0.232	0.82
AstCalc (sqmm)	253.35	327.08	384.56	803.91	197.13	697.5
Ast Prv (sqmm)	603.18	603.18	603.18	942.48	942.48	942.48
Reinforcement	3-T16	3-T16	3-T16	3-T20	3-T20	3-T20

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	85.27	55.31	76.87
Asv Torsion (sqmm/m)	301.82	206.99	276.76
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 140

SFR : -

Beam No		:	**B28**	
Group No		:	**G9**	
Analysis Reference(Member)	5.029 m	:	169	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	15.37	15.37	15.37	26.21	41.7	53.16
PtClc (%)	0.2	0.2	0.2	0.2	0.296	0.386
AstCalc (sqmm)	170.2	170.2	170.2	170.2	252.06	328.26
Ast Prv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	68.38	77.99	83.47

Asv Torsion (sqmm/m)	434.35	478.64	495.61
Asv Reqd (sqmm/m)	254.81	300.21	341.25
Asv Prv (sqmm/m)	718.14	718.14	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 140	2L-T8 @ 140

SFR : -

Beam No		:	**B29**	
Group No		:	**G9**	
Analysis Reference(Member)	5.029 m	:	170	
Breadth		:	230	mm
Depth		:	430	mm
Concrete Grade		:	M25	N/sqmm
Grade Of Steel (Main)		:	Fe550	N/sqmm
Grade Of Steel (Shear)		:	Fe550	N/sqmm
Top/Bottom Clear Cover		:	30	mm
Side Clear Cover		:	30	mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type		:	Regular Beam	

Flexure Design

	Beam Bottom			**Beam Top**		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	13.35	13.35	3.27	45.61	21.12	22.11
PtClc (%)	0.2	0.2	0.2	0.326	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	277.63	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	32.92	16.02	33.01
Asv Torsion (sqmm/m)	126.46	54.5	150.12
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		:	**B30**
Group No		:	**G9**
Analysis Reference(Member)	5.029 m	:	171

Breadth	:	230	mm
Depth	:	430	mm
Concrete Grade	:	M25	N/sqmm
Grade Of Steel (Main)	:	Fe550	N/sqmm
Grade Of Steel (Shear)	:	Fe550	N/sqmm
Top/Bottom Clear Cover	:	30	mm
Side Clear Cover	:	30	mm
Design Code	:	IS 456 : 2000 + IS 13920 : 2016	
Beam Type	:	Regular Beam	

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	6.2	14.63	8.33	21.83	1.78	21.62
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	30.64	15.76	30.73
Asv Torsion (sqmm/m)	114.18	76.13	114.48
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	365.6	365.6	365.6
Reinforcement	2L-T8 @ 275	2L-T8 @ 275	2L-T8 @ 275

SFR : -

Beam No		**:**	**B31**
Group No		**:**	**G9**
Analysis Reference(Member)	5.029 m	:	172
Breadth		:	230 mm
Depth		:	430 mm
Concrete Grade		:	M25 N/sqmm
Grade Of Steel (Main)		:	Fe550 N/sqmm
Grade Of Steel (Shear)		:	Fe550 N/sqmm
Top/Bottom Clear Cover		:	30 mm
Side Clear Cover		:	30 mm
Design Code		:	IS 456 : 2000 + IS 13920 : 2016
Beam Type		:	Regular Beam

Flexure Design

	Beam Bottom			Beam Top		
	Left	**Mid**	**Right**	**Left**	**Mid**	**Right**
Mud (kNm)	10.89	24.38	22.59	27.13	4.43	18.53
PtClc (%)	0.2	0.2	0.2	0.2	0.2	0.2
AstCalc (sqmm)	170.2	170.2	170.2	170.2	170.2	170.2
Ast Prv (sqmm)	226.2	226.2	226.2	339.3	339.3	339.3
Reinforcement	2-T12	2-T12	2-T12	3-T12	3-T12	3-T12

Shear Design

	Left	**Mid**	**Right**
Vut (kN)	47.33	28.89	39.92
Asv Torsion (sqmm/m)	218.45	151.86	195.53
Asv Reqd (sqmm/m)	254.81	254.81	254.81
Asv Prv (sqmm/m)	718.14	365.6	718.14
Reinforcement	2L-T8 @ 140	2L-T8 @ 275	2L-T8 @ 140

SFR : -